CSC 380 — Quiz 04 Practice Guide

Banner — PUBLISHED SCOPE; due date NEEDS VERIFICATION.

- **Scope is published** (D2L announcement, Sep 18, 2026): Bayes' rule, the law of total probability (LOTP), and independence, covering **slides pp. 43–68**.
- **The due date is not verified.** Quiz 04 was not on Gradescope as of the 2026-09-24 audit (~14:03 Arizona time). The syllabus cadence ("quizzes every Tuesday unless noted") is **not** a verified due date. **Never invent a due date.** This is an open practice guide, not a countdown plan.

Canonical maps: `csc380-topic-map.md` · `announced-assessments-inventory-2026-09-24.md`

1. Header

Field	Value	Status
Course	CSC 380 Principles of Data Science (FA26), TuTh 5:00–6:15 PM, M Pacheco ILC 130	Confirmed
Assessment	Quiz 04 (practice guide)	Item announced
Due date/time	**Needs verification** — do not invent	**NV**
Coverage	Bayes' rule; LOTP; independence; slides 43–68	**PUBLISHED SCOPE**
Quiz policy (syllabus)	12 quizzes, each 1.5% of the grade; the two lowest are dropped; no make-ups	Confirmed (syllabus)
Past quiz format	Quiz 01 was **one** graded rubric question on a **one-page** submission, worth 1.5 points	Practice recommendation (Gradescope notes)
Guide updated	2026-09-24 (America/Phoenix) — Opus review pass	

All times are **Arizona time (MST)**.

Related but separate: HW3 (Probability 2) is due **Tue Sep 29, 11:59 PM**. It has **two** Gradescope items, HW03 (written) and HW03 Code, and both must be submitted. It overlaps these topics, but HW3 is **not** this guide's target, and nothing here solves it.

Notation. In this guide, **not-A** means the complement of A (often written A^c or $\bar{A}$ on slides). P(A | B) is read "probability of A given B."

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2. Topic checklist (learning order)

Order	Section ID	Topic	Blind ✓	Check ✓	Retake ✓
1	`380-Q04-refresh`	Conditional probability refresher			
2	`380-Q04-lotp`	Law of total probability			
3	`380-Q04-bayes`	Bayes' rule			
4	`380-Q04-indep`	Independence (vs. mutually exclusive)			
5	`380-Q04-slides`	Work the examples on slides 43–68			

Alias: `380-L-prob3` ≈ `380-Q04-bayes`, for calendar continuity.

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3. Plain explanations + worked examples

Conditional probability refresher (`380-Q04-refresh`)

P(A | B) = P(A and B) / P(B), when P(B) > 0.

Rearranged, this gives the **multiplication rule**: P(A and B) = P(A | B)·P(B).

The Gradescope scores show the week-2 quiz (conditional probability) was the weakest of the three, so this refresher comes first.

Law of total probability (`380-Q04-lotp`)

If B_1, ..., B_k **partition** the sample space (they don't overlap and together cover everything):

$P(A) = \Sigma_i\ P(A \mid B_i) \cdot P(B_i)$

With two parts: P(A) = P(A | B)·P(B) + P(A | not-B)·P(not-B).

The intuition: split the world into cases, find A's probability in each case, and weight each case by how likely it is.

Bayes' rule (`380-Q04-bayes`)

$P(B_j \mid A) = P(A \mid B_j) \cdot P(B_j) / P(A)$, where P(A) comes from LOTP.

Table method (it makes normalizing automatic):

Hypothesis	Prior	Likelihood P(data given H)	Prior × likelihood	Posterior
...	...	...	...	(row value) ÷ (column sum)

Worked example (disease test). P(D) = 0.01, P(+ | D) = 0.9, and P(+ | not-D) = 0.05.

- P(+) = 0.9·0.01 + 0.05·0.99 = 0.009 + 0.0495 = 0.0585.
- P(D | +) = 0.009 / 0.0585 ≈ **0.154**.

The disease is rare (a low **base rate**), so most positive results are false positives, even though the test is fairly accurate.

Independence (`380-Q04-indep`)

A and B are independent when **P(A and B) = P(A)·P(B)**. Equivalently, P(A | B) = P(A) (when P(B) > 0), or P(B | A) = P(B).

- **Coin example.** For two fair flips, $P(H_1 \text{ and } H_2) = 1/4 = (1/2)(1/2)$, so the flips are independent.
- **Mutually exclusive is not independent.** If P(A) > 0 and P(B) > 0 but they cannot both happen, then P(A and B) = 0 ≠ P(A)·P(B). Knowing A happened tells you B did not, which is maximal dependence.
- To **check** independence, compute both sides of the product rule. Don't rely on intuition.

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4. Practice questions (core)

Recall

- R1. Write Bayes' rule for hypotheses H and not-H, given data D.
- R2. State independence in three equivalent ways.

Solve

- S1. Three urns are equally likely: U1 has 2 red and 1 blue, U2 has 1 red and 1 blue, U3 has 0 red and 2 blue. You pick an urn at random and draw one ball, which is red. What is P(U1 | red)?
- S2. P(A) = 0.3, P(B) = 0.4, P(A and B) = 0.12. Are A and B independent?
- S3. P(A | B) = 0.5, P(B) = 0.2, P(A | not-B) = 0.1. Find P(A) and P(B | A).

Trace

- T1. For S1, fill in the Bayes table: prior, likelihood, prior × likelihood, posterior.

Explain

- E1. Why does LOTP appear in the denominator of Bayes' rule?
- E2. What is the difference between independent and mutually exclusive?

Code (optional study aid)

- C1. Simulate about 10,000 draws of the S1 setup in Python. Compare the fraction of red draws that came from U1 with the exact answer.

4b. One-page mock quizzes (1.5 points each) — Practice recommendation

Where this format comes from. Quiz 01's Gradescope result was one rubric question on a one-page submission, worth 1.5 points. Each mock below is **one multi-part problem that fits on one page**, with three parts worth 0.5 points each. The Quiz 04 interface and due date are **not known**. Give each mock **10 minutes**, closed-notes.

Mock Quiz A — LOTP + Bayes (factory)

Machine M1 makes 60% of the items, and 2% of its items are defective. Machine M2 makes the other 40%, and 5% of its items are defective.

- (a) [0.5] Find P(defective).
- (b) [0.5] An item is defective. Find P(M2 | defective).
- (c) [0.5] In one sentence, explain why (b) is larger than 0.4.

Mock Quiz B — Independence (dice)

Roll two fair six-sided dice. Let A = "the first die shows 6", B = "the sum is 7", and C = "the sum is 12".

- (a) [0.5] Are A and B independent? Show the product check.
- (b) [0.5] Are A and C independent? Show the product check.
- (c) [0.5] P(E) = 0.2, P(F) = 0.5, and E and F are mutually exclusive. Are they independent?

Mock Quiz C — Table → conditional → independence

Of 200 students, 80 drink coffee. 50 of the coffee drinkers are night owls, and 40 of the non-coffee drinkers are night owls.

- (a) [0.5] Find P(night owl).
- (b) [0.5] Find P(coffee | night owl).
- (c) [0.5] Are "coffee" and "night owl" independent? Justify with numbers.

Mock Quiz D — Bayes (spam filter)

P(spam) = 0.3. The word "free" appears in 40% of spam emails and in 5% of non-spam emails.

- (a) [0.5] Find P("free").
- (b) [0.5] Find P(spam | "free").
- (c) [0.5] Explain in one sentence the mistake in saying "P(spam | free) = 0.4".

Quick checks

- **Qμ1.** P(A) = 0.5, P(B) = 0.4, P(A and B) = 0.1. Independent? Give a one-line reason.
- **Qμ2.** Write LOTP for an event D with the partition {B, not-B}.
- **Qμ3.** P(H) = 0.2, P(E | H) = 0.9, P(E | not-H) = 0.3. Find P(H | E), showing the LOTP denominator.
- **Qμ4.** True or false: mutually exclusive events with positive probabilities are independent.
- **Qμ5 (stretch).** Using the disease test from §3, a second, independent test is also positive. Assume the two results are independent given disease status. Find P(D | both positive).

5. ANSWER KEY (separate — check after a blind attempt)

Core keys

R1. P(H | D) = P(D | H)·P(H) / [P(D | H)·P(H) + P(D | not-H)·P(not-H)]

R2. P(A and B) = P(A)·P(B); P(A | B) = P(A); P(B | A) = P(B). The last two require the conditioning event to have positive probability.

S1. P(red) = (2/3)(1/3) + (1/2)(1/3) + 0 = 2/9 + 1/6 = 7/18. So P(U1 | red) = (2/9) / (7/18) = **4/7**.

S2. 0.3 × 0.4 = 0.12, which equals P(A and B). **Yes, independent.**

S3. P(A) = 0.5·0.2 + 0.1·0.8 = **0.18**. P(B | A) = 0.10 / 0.18 = **5/9**.

T1.

Urn	Prior	Likelihood	Prior × likelihood	Posterior
U1	1/3	2/3	2/9 = 4/18	4/7
U2	1/3	1/2	1/6 = 3/18	3/7
U3	1/3	0	0	0
Sum	1		7/18	1

E1. The denominator is P(data), the total probability of the data across all hypotheses. LOTP computes exactly that, and dividing by it makes the posteriors sum to 1.

E2. Mutually exclusive means P(A and B) = 0. Independent means P(A and B) = P(A)·P(B), which is usually greater than 0. With positive probabilities, the two properties cannot both hold.

C1. The empirical fraction should land near 4/7 ≈ 0.571, within about ±0.02 for 10,000 draws.

Mock quiz keys

Quiz A.

- (a) 0.6·0.02 + 0.4·0.05 = 0.012 + 0.020 = **0.032**.
- (b) 0.020 / 0.032 = **0.625**.
- (c) M2 has the higher defect rate, so seeing a defect shifts belief toward M2, from the prior of 0.4 up to 0.625.

Quiz B.

- (a) P(A) = 1/6. P(B) = 6/36 = 1/6. P(A and B) = P(first die 6, second die 1) = 1/36, and (1/6)(1/6) = 1/36. **Independent.**
- (b) P(C) = 1/36. P(A and C) = 1/36, but P(A)·P(C) = 1/216. **Not independent.**
- (c) P(E and F) = 0, but P(E)·P(F) = 0.1. **Not independent.**

Quiz C.

- (a) (50 + 40) / 200 = **0.45**.
- (b) 50 / 90 = **5/9 ≈ 0.556**.
- (c) P(owl | coffee) = 50/80 = 0.625, which is not P(owl) = 0.45. Equivalently, P(coffee and owl) = 0.25, but P(coffee)·P(owl) = 0.4·0.45 = 0.18. **Not independent.**

Quiz D.

- (a) 0.4·0.3 + 0.05·0.7 = 0.12 + 0.035 = **0.155**.
- (b) 0.12 / 0.155 ≈ **0.774**.
- (c) 0.4 is P(free | spam), the likelihood. Treating it as P(spam | free), the posterior, ignores the prior and the base rate.

Quick-check keys

- **Qμ1.** No. 0.5·0.4 = 0.2 ≠ 0.1.
- **Qμ2.** P(D) = P(D | B)·P(B) + P(D | not-B)·P(not-B).
- **Qμ3.** P(E) = 0.9·0.2 + 0.3·0.8 = 0.42. P(H | E) = 0.18 / 0.42 = **3/7 ≈ 0.429**.
- **Qμ4.** False. Their intersection has probability 0, but the product of their probabilities is greater than 0.
- **Qμ5.** 0.81·0.01 / (0.81·0.01 + 0.0025·0.99) = 0.0081 / 0.010575 ≈ **0.766**. Equivalently, use 0.154 as the new prior and update once more.

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6. Common mistakes + readiness checklist

- Confusing P(A | B) with P(B | A) (base-rate neglect).
- Forgetting to normalize, i.e. skipping the LOTP denominator.
- Treating mutually exclusive events as independent.
- Using a "partition" whose parts overlap or leave gaps.
- **Inventing a Quiz 04 due date** from the Tuesday cadence.

Ready if:

- [ ] Mock Quizzes A–D are each done in 10 minutes or less with at least 1.0 of 1.5 points
- [ ] Three independence checks are done by computing both sides
- [ ] The examples on slides 43–68 are worked
- [ ] **No countdown cram** until the due date is verified

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7. Open practice checklist (no due-date cram plan)

Mode	Action
Anytime	Rotate short sets for `380-Q04-refresh`, `lotp`, `bayes`, and `indep`
Alongside HW3 (due Sep 29, 11:59 PM)	Work these topics as separate practice. Upload **both** HW03 and HW03 Code.
When the due date is posted on Gradescope or D2L	Add a light review 24–48 hours before. **Do not invent the date now.**
Tuesdays and Thursdays	Keep light around classes. Do deep drills Mon/Wed/Fri and weekends.

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8. Changes in this review (2026-09-24)

- Kept the **PUBLISHED SCOPE** and **due-date NV** banners unchanged in meaning.
- Replaced the complement symbol "^{c}" with "not-A", because it rendered as a missing-glyph box in the old PDF. Added a notation note.
- Added quiz policy facts from the syllabus (12 quizzes, drop the 2 lowest, no make-ups) and the HW3 two-item submission reminder.
- Added Mock Quizzes A–D (one page, three parts at 0.5 points each), a filled Bayes table (T1), and a stretch sequential-update item. All have keys.
- Stated the weak-early-quiz signal in words only, with no scores, because the repository is public.
- Changed times to Arizona time.

9. Study-aid / AI-policy disclaimer

Study aid only — not for submitted work (Quiz 04, HW, exams). Follow CSC 380 and UA academic-integrity and AI policies.